

Kanwarpal Brar

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Education

University of Waterloo

Waterloo, Ontario, Canada

Bachelor of Computer Science (BCS) | GPA: 3.84/4.0

2020 - 2025

Courses: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Distributed Systems, Concurrency, Databases

Work Experience

Meta

Production Engineer

Sept 2025 - July 2026

- Accelerated **Meta Blob Storage** turnup **95%** via **Python** automation over **Thrift** schemas, provisioning capacity for **AI training** fleets
- Saved **100+** engineering hours by shipping a **C++ key-value** state layer atop **Infrastructure-as-Code** definitions, onboarding legacy systems
- Restored cluster capacity in **24 hours** by diagnosing cross-stack **garbage collection SEV**, unblocking environments for **6** downstream services
- Safeguarded **99% SLO** across **7** MBS services as **oncall**, driving **observability** and **incident response** for **Meta Superintelligence Labs**
- Scaled team throughput **3x** via **LangChain AI agent** pipelines monitoring infrastructure, cutting change cycles from **hours to minutes**

Carta

Payments Software Engineering Co-op

Sept 2024 - Dec 2024

- Averted **\$5000+** in regulatory penalties within first **2 weeks** by implementing financial compliance controls with **Django + React**
- Managed **\$5M+** monthly leading international banking integration expansion, leveraging **Python, gRPC**, and **Domain Driven Design**
- Architected fault-tolerant **microservices** on **AWS** with **Docker** and **Kubernetes**, achieving **99.99%** uptime across distributed services
- Slashed network overhead **50%** by implementing centralized money movement controls with **RBAC** in a **Java Microservice**
- Accelerated feature development **40%** through flexible Django permission system using YAML-templated definitions

University of Waterloo

Distributed Systems Research Assistant

May 2024 - Aug 2024

- Benchmarked **serverless** frameworks with **WRK** and custom **Bash** scripts, informing design decisions for high-performance system architecture
- Delivered **200+** requests/second per node by deploying & optimizing **Kubernetes** clusters with **KNative** Serving and **Istio** ingress controls
- Cut cold-start latency **20%** through targeted autoscaling and **TCP/IP networking** optimization, increasing platform responsiveness
- Designed architectural changes improving resource utilization **15%**, validated with monitoring scripts in a comprehensive technical report

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Backend Software Engineering Co-op

Jan 2024 - Apr 2024

- Boosted OCX report generation speed **10%** using an $O(n)$ complexity cell management system with **Apache POI + Java**, streamlining results
- Improved semantic search relevance by **6%** via refined prompt engineering for report queries & real-world use-cases
- Eliminated **3000+** tickets by streamlining Ownership Report permissions in **Django**, collaborating cross-functionally with product leaders
- Slashed user wait times by **10%** through optimizing ownership report queries in **Spring**, enhancing system responsiveness

Arctic Wolf Networks

Software Developer Co-op

May 2023 - Aug 2023

- Reduced response times **25%** by implementing a concurrent **Go** monitoring system for **Prometheus** metrics on an **Apache Kafka** wrapper
- Attained **100%** accuracy identifying regression-test gaps with a reflection-based **Go** unit test verifier for an **Apache Kafka** wrapper
- Expedited development by creating a forward/backward compatible Kafka Serialization system using **SchemaVer** and **Avro**

Genesys Cloud Services

Full Stack Software Developer Co-op

Sept 2022 - Dec 2022

- Lowered hosting costs by **5%** by migrating a monolithic scheduling API to an end-to-end serverless architecture with **Python + Flask + Lambda**
- Compressed codebase size by **30%** and enhanced maintainability through a documented **Python + Flask + OpenAPI** service for scheduling

Projects

Secure Self-Hosted Kubernetes Platform 🔒

2025

- Configured the **Linux OS** and host networking for a **bare-metal Kubernetes** cluster with **Traefik** TCP/UDP and HTTPS ingress
- Automated deployments with **GitHub Actions** and declarative manifests, using readiness and liveness checks for zero-downtime rollouts
- Added **Prometheus** alerting and **Grafana** dashboards, with **Cloudflare Tunnels** and **WARP** securing remote administration

Distributed File System 🗄️

2025

- Designed a **C++** distributed file system using **FUSE**, custom **RPC**, and threading for metadata and data operations
- Implemented a lease-based client cache, reducing network calls by **60%** and average read latency by **30%**

Skills

Languages: Go, Python, Bash, C++, Java, TypeScript, JavaScript, C, C#, Swift, SQL

AI / GenAI: LLMs, RAG, Prompt Engineering, LLM Evals, AI Agents, Agentic Workflows, Claude Code, MCP, Hermes, OpenClaw

Frameworks: Spring Boot, Django, CUDA, Gin, Node.js, Express.js, React, React Native, GraphQL, REST APIs, gRPC, Metal, Xcode

Infrastructure: Kubernetes, Prometheus, Grafana, Linux, Networking, Distributed Systems, HPC, AWS, Docker, Terraform, CI/CD, Git, eBPF

Data: PostgreSQL, MongoDB, Redis, Elasticsearch, Kafka, Spark, HBase, Protocol Buffers, Avro, Thrift